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Preference for and acceptability of a fat-microencapsulated anionic salt by dry dairy cows

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ABSTRACT

The objectives of this study were to determine (1) the preference for, and acceptability of, a novel, fat-microencapsulated anionic salt by nonlactating dairy cows and (2) whether these responses were influenced by cow personality. Twenty-four pregnant Holstein dairy cows (parity at dry-off: median = 2; range = 1–5) were enrolled at dry-off (56 d to calving), in cohorts of 6, and housed in a tiestall facility. After adapting to a base dry cow diet for 14 d, cows were exposed to each of 2 treatment diets in a crossover design (with 7-d periods): (1) base dry cow diet with added fat-microencapsulated anionic salt (ANIC) at 0.30 kg/cow/d DM (providing −2,544 mEq/d at that dosage; formulated DCAD = −101.5 mEq/kg of expected DMI), or (2) base dry cow diet with fat placebo (PLCB) at 0.15 kg/cow per day DM (formulated DCAD = +64 mEq/kg of expected DMI). Following the crossover design, cows were offered both diets simultaneously in a 3-d preference test; feed was offered in a split bunk, in large enough quantities to ensure ad libitum daily access to both diets. Following the preference test cows were assessed for personality traits using a combined arena test comprised 3 parts: a novel arena test, a novel object test, and a novel human test. Dry matter intake was calculated for each day, and urine pH was measured 2×/wk for every cow throughout the study. When cows consumed the ANIC diet, they had lower urine pH values than when on the PLCB diet (7.6 vs. 8.2). No difference in DMI was detected between the ANIC and PLCB diets (19.3 vs. 19.5 kg/d DM). During the preference phase, cows chose to eat a similar proportion of their total daily DMI for the ANIC and PLCB diets (48.6% vs. 51.4% of total DMI). Cows sorted against long particles of the PLCB diet and against fine particles of both diets during the acceptability phase. During the preference test, cows sorted against the longest ration particles of the PLCB diet. Principal

component analysis of the behaviors recorded during the personality assessment revealed 2 factors interpreted as personality traits, “active” and “bold.” Less bold cows consumed a higher portion of the ANIC diet during the preference test. Less active cows sorted more for the medium particles of both diets, whereas more active cows sorted more for the short particles of both diets during the preference test. Overall, the results demonstrate that dry cows showed no preference between a TMR containing a fat-microencapsulated anionic salt or a fat placebo, and demonstrated similar acceptability of both diets. Further, feed sorting and diet preference were influenced by cow personality.

Key words: preference test, anionic salt, personality, palatability

INTRODUCTION

Hypocalcemia, both in its clinical and subclinical form, may be prevalent in postpartum dairy cows and is associated with greater incidence of other postpartum diseases (McArt and Neves, 2020), reduced reproductive performance (Venjakob et al., 2018), and greater risk of culling (Venjakob et al., 2018; McArt and Neves, 2020). One strategy to help prevent hypocalcemia is the feeding of acidogenic diets (i.e., diets with a negative DCAD) during the close-up period, which induce a compensated metabolic acidosis characterized by increased hydrogen ion concentration in the systemic circulation, resulting in an abnormally low serum bicarbonate level (Charbonneau et al., 2006). These systemic effects lead to greater parathyroid hormone receptiveness, resulting in increased production of calcitriol, which in turn enhances bone resorption and intestinal absorption of Ca (Charbonneau et al., 2006), which helps prevent the decline in blood Ca concentrations after calving (Santos et al., 2019; Wilkens et al., 2020; Melendez and Chelikani, 2022). These negative DCAD diets can be formulated through the inclusion of anionic salts (Santos et al., 2019).

Achieving a sufficient degree of compensated metabolic acidosis without reducing DMI can be difficult

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(Charbonneau et al., 2006; Melendez and Chelikani, 2022). Poor palatability of salts (Oetzel, 1988) and the systemic effects associated with the induced metabolic acidosis (Vagnoni and Oetzel, 1998) were previously proposed as potential explanations for the reduced prepartum DMI of cows fed negative DCAD diets. Zimpel et al. (2018) more recently demonstrated that it is more likely the metabolic acidosis induced by negative DCAD, and not necessarily the anions themselves, that may lead to DMI depression. Regardless, depression in DMI may still occur with acidogenic diets; meta-analyses indicate that although anionic salt source can affect magnitude of the prepartum decrease in DMI, this reduction in DMI appears to occur regardless of the source (Charbonneau et al., 2006; Santos et al., 2019). Because adequate and consistent DMI is needed to ensure sufficient intake of anionic salts to achieve desired levels of metabolic acidosis, continued efforts must be placed on refining DCAD diet formulations, particularly their included anionic salts, to minimize any effects on DMI. One possibility is to bind or encapsulate the anionic salts with other feed ingredients that are more palatable. Certain fats can be quite palatable to dairy cows and able to mask the flavor of less palatable ingredients. Hultquist and Casper (2016) demonstrated that encapsulating potassium carbonate with free fatty acids (a mixture of palmitic, stearic, oleic, and myristic acids) did not create any palatability issues when fed to dairy cows.

Palatability, and associated diet selection (feed sorting), are known to be highly variable among individual dairy cows (Leonardi and Armentano, 2003; Melin et al., 2005; Neave et al., 2018). Dairy cows typically preferentially select (sort) for the more palatable concentrate particles in a TMR, and against the longer forage components, whereas some researchers have demonstrated that cows are variable in this and sort in an opposite fashion (Leonardi and Armentano, 2003). One possible explanation for this variability in palatability and preference for certain feeds is individual cow personality (Meagher et al., 2017). Dairy cow personality can be defined as individual differences in behavior that are consistent across time and situations (Finkemeier et al., 2018). Personality in dairy cows is associated with performance and welfare, as it influences how animals respond to different management practices and situations (Finkemeier et al., 2018). It has been demonstrated that personality influences feeding behavior when cows are faced with changing environmental and social conditions (Schwanke et al., 2022; 2024). Personality traits such as boldness and exploration have been linked to variability in feeding behaviors, with bolder, and more explorative dairy cows more likely to consume novel feeds, and are less reactive with novelty at the feed bunk (Meagher et al., 2017; Neave et al., 2018). Personality assessment may, thus,

be used as a tool to understand individual variability in behavioral responses to dietary interventions (Neave et al., 2018).

Thus, the objectives of this study were to determine (1) the preference for, and acceptability of, a fat-microencapsulated anionic salt by nonlactating dairy cows and (2) whether these responses are influenced by cow personality. It was hypothesized that when presented with both dry cow TMR with a fat-microencapsulated anionic salt and one with a fat placebo, cows will have no difference in preference, and there will be no difference in DMI when offered the feed types on their own. It was also predicted that certain personality traits, such as boldness and exploration, may influence the preference of the cows and degree of feed sorting of the test diets.

MATERIALS AND METHODS

Animals, Housing, and Experimental Design

Twenty-four pregnant Holstein cows completing their first or greater lactation (parity median = 2; range = 1 to 5; mean = 1.8 ± 0.3 ; mean $\pm$ SD) were dried off 66.2 $\pm$ 1.6 d before parturition and enrolled in this study. Cows were housed at the University of Guelph Elora Research Station, Ontario Dairy Research Centre (Elora, ON, Canada), which took place from September 2023 to February 2024. For dry-off, cows were moved from their lactating pens to individual tiestalls (1.2 $\times$ 1.8 m; width $\times$ length) with a mattress base and bedded with chopped straw. Stalls were bedded daily, as needed, and cleaned 3 $\times$ /d at 0600 h, 1130 h, and 1800 h. Each stall had individual access to feed and ad libitum access to a water bowl. All cow use and experimental procedures used throughout this trial complied with the guidelines of the Canadian Council on Animal Care (2009) and were approved by the University of Guelph Animal Care Committee (protocol #4903).

Sample size and power analysis were completed (as per Morris, 1999) to calculate the minimum number of replicates required per treatment. To detect a 4.3% difference in DMI between treatments with an average coefficient of variation of 7.5% for DMI (Havekes et al., 2020a,b), we calculated the minimum number of replicates per treatment to be 24.

Cows ($n = 24$) were enrolled sequentially in 4 cohorts of 6, by selecting cows with similar estimated calving dates. Upon dry-off, cows were acclimated to the base dry cow TMR (Table 1) for 1 wk before being offered either treatment diet. Initially, during the acceptability test period at 52.2 ± 1.6 d to calving, cows were exposed to each of 2 treatment diets in a crossover design, with 7 d on each treatment, with half of the cows assigned to each treatment to start: (1) base diet with a propri-

etary fat microencapsulated anionic salt added at 0.30 kg/d DM per cow (formulated DCAD = -101.5 mEq/kg of expected DMI; 49% fat; Jebo Nutrition Inc., Saint-Hyacinthe, QC, Canada; **ANIC**), and (2) base diet with added fat placebo at 0.15 kg/d DM per cow (formulated DCAD = $+64$ mEq/kg of expected DMI; 98% fat; Jebo Nutrition Inc., Saint-Hyacinthe, QC, Canada; **PLCB**). The anion salt product provided $-2,544$ mEq/d at the supplement amount per cow per day mentioned above. Following this, cows were offered both diets at once (as described below) in a 3-d preference test at 38.2 ± 1.6 d to calving. Both diets were fed ad libitum. The base dry cow ration consisted of wheat straw (chopped to 2.54 cm in length by a Haybuster Model H-1150, Jamestown, ND), corn silage, alfalfa haylage, and a dry cow supplement (Table 1), which was prepared in a mixer wagon (Jaylor Model 5572, Jaylor Fabricating, Orton, ON, Canada) then transferred into individual feed carts (Super Data Ranger, American Calan, Northwood, NH). Both additives were individually added to each feed cart and mixed for 5 min before feeding. The base dry cow TMR was formulated to meet 100% the nutrient requirements for a 700-kg dry cow (NASEM, 2021), with an expected DMI of 15 kg/d, and formulated using AMTS.Cattle.Professional 4.10.5 (Agricultural Modeling and Training Systems, Groton, NY). The amount of feed offered was adjusted daily to target a refusal rate of 5% to 10% (actual: $11.9\% \pm 0.3\%$ DM). Fresh feed was delivered twice daily, with a first delivery at 1100 h, where cows were offered 60% of their daily allotment and the remaining 40% was offered at 1400 h.

Throughout the study, cows had their feed provided in 2 experimental bins with a split-bunk divider (bin: $72 \times 62 \times 23$ cm; divider: $72 \times 2 \times 60$ cm; length $\times$ width $\times$ height) were placed in the feed bunk area of each stall (Figure 1). During the acceptability phase, where cows were offered 1 of the 2 treatment diets in a crossover design, each diet was administered to either the left or right bin. For each cow, a specific diet was only administered to one side, with the other diet being offered to the other side, which was balanced among all cows. During the preference test, both treatment diets were administered simultaneously to the same bin in which cows previously received each treatment diet, with sufficient daily amounts of each diet so that they had the opportunity, if they chose, to consume all of their daily intake from only one of the diets (orts = $33.6\% \pm 1.7\%$ DM). Following the preference test, all cows were returned to the standard freestall, dry cow housing at the research barn.

Feed Intake, Feed Sampling and Analysis, and Feed Sorting

Daily individual feed intake was calculated using the weight of the feed offered and the weight of the refusals, corrected to DMI (kg/d), using the DM percentage calculated from taken feed samples. During the preference test, the proportion of daily intake for each treatment diet was calculated by dividing the DMI of each diet by the total DMI for each day. Fresh feed samples for the base and treatment diets were collected 3 $\times$ /wk, in duplicate, and ration ingredients were sampled every 3 wk, in duplicate, for the entirety of the trial, to determine DM and chemical composition, as well as particle size distribution. Orts (refusal) samples from each cow were collected 3 $\times$ /wk, in duplicate, at the end of each experimental day (i.e., the next morning before feed refusal cleanup). During the preference test, fresh feed samples were taken in duplicate on d 1 and 3, and orts samples were at the end of each experimental day (i.e., the next morning before feed refusal cleanup). All feed samples were stored at -20°C upon collection. Feed samples were thawed for 24 h before being oven-dried at 60°C for 48 h for DM analysis. Samples taken for particle size distribution, including dry TMR, forage samples, and orts samples were separated, before drying, into 4 fractions: long (>19 mm), medium (8–19 mm), short (4 to 8 mm), and fine (<4 mm) using a Penn State Particle Separator (**PSPS**; Table 2; Heinrichs, 2013). Once dried, samples were ground to pass through a 1-mm screen (Model 4 Wiley Laboratory Mill, Thomas Scientific, Swedesboro, NJ). Ground samples were pooled weekly for treatment rations and PSPS samples and every 6 wk for ration ingredients and sent to A&L Laboratory Services Inc. (London, ON, Canada) for analysis of ash (550°C ; method 942.05; AOAC International, 2000), ADF (method 973.18; AOAC International, 2000), NDF with heat-stable α -amylase and sodium sulfite (method 2002.04; AOAC International, 2000), CP ($\text{N} \times 6.25$; method 990.03; AOAC International, 2000), Leco FP-628 Nitrogen Analyzer, Leco Corp., St. Joseph, MI), starch (heat-stable amylase and amyloglucosidase; method 996.11; AOAC International, 2000), sugar (method 968.28; AOAC International, 2000), crude fat (using pet ether; method 920.39; AOAC International, 2000), lignin (using ADF residue and H_2SO_4), minerals (using aquaregia digestion inductively coupled plasma atomic emission spectroscopy), and calculation of net energy (using NRC [2001] equations).

Sorting for each PSPS fraction was determined by dividing the actual consumption of each fraction by the predicted consumption of that respective fraction, expressed as a percentage (Leonardi and Armentano, 2003). Actual consumption for each fraction of the TMR was estimated

Table 1. Ingredient and chemical composition (mean $\pm$ SD) of the base dry cow diet and treatment diets

Item	Base diet	Treatment diet	
		ANIC ¹	PLCB ²
Ingredient (% of DM)			
Wheat straw ³	26.6	26.1	26.3
Corn silage ⁴	34.3	33.8	33.9
Alfalfa haylage ⁵	15.2	15.0	15.1
Dry supplement ⁶	23.9	23.6	23.7
Anionic salt ⁷	—	1.5	—
Fat placebo	—	—	1.0
Chemical composition			
DM (%)	54.0 ± 0.56	54.5 ± 0.22	54.0 ± 0.39
CP (% of DM)	11.8 ± 0.43	12.2 ± 0.24	11.2 ± 0.28
ADF (% of DM)	31.4 ± 0.57	31.3 ± 0.42	32.4 ± 0.54
NDF (% of DM)	47.2 ± 0.50	47.0 ± 0.93	48.2 ± 0.61
Starch (% of DM)	14.1 ± 0.86	14.5 ± 0.28	14.6 ± 0.31
Sugar (% of DM)	1.01 ± 0.37	1.13 ± 0.40	0.75 ± 0.14
NEL (Mcal/kg DM)	1.47 ± 0.01	1.47 ± 0.01	1.44 ± 0.01
Crude fat (% of DM)	2.42 ± 0.04	3.47 ± 0.08	3.51 ± 0.05
Ash (% of DM)	7.12 ± 0.24	7.27 ± 0.19	6.87 ± 0.19
Ca (% of DM)	1.33 ± 0.08	1.13 ± 0.04	1.15 ± 0.03
P (% of DM)	0.50 ± 0.02	0.44 ± 0.02	0.43 ± 0.01
K (% of DM)	1.31 ± 0.03	1.24 ± 0.04	1.29 ± 0.03
Mg (% of DM)	0.41 ± 0.02	0.45 ± 0.02	0.39 ± 0.02
Na (% of DM)	0.16 ± 0.01	0.14 ± 0.01	0.13 ± 0.01
S (% of DM)	0.22 ± 0.01	0.34 ± 0.01	0.21 ± 0.01
Cl (% of DM)	0.34 ± 0.02	0.67 ± 0.02	0.31 ± 0.02
DCAD (mEq/kg) ⁸	172.2 ± 3.70	-22.4 ± 8.36	168.7 ± 8.11

¹Base diet with added fat-microencapsulated anionic salt (Jefo Nutrition Inc., Saint-Hyacinthe, QC, Canada) at 0.30 kg/d DM per cow (formulated DCAD = -101.5 mEq/kg DM; 49% fat).

²Base diet with added fat placebo (Jefo Nutrition Inc., Saint-Hyacinthe, QC, Canada) at 0.15 kg/d DM per cow (formulated DCAD = +64 mEq/kg DM; 98% fat).

³Wheat straw had a DM of 92.4% and chemical composition (DM basis) of 2.4% $\pm$ 0.19% CP, 0.9% $\pm$ 0.20% starch, 53.7% $\pm$ 1.14% ADF, and 84.7% $\pm$ 1.03% NDF. Particle size distribution (% of DM) = 43.2% $\pm$ 5.5% long (>19mm), 22.8% $\pm$ 5.9% medium (8 to 19 mm), 16.3% $\pm$ 0.6% short (4 to 8 mm), and 17.7% $\pm$ 1.0% fine (<4 mm) particles.

⁴Corn silage had a DM of 37.5% and chemical composition (DM basis) of 7.4% $\pm$ 0.34% CP, 35.0% $\pm$ 1.36% starch, 19.3% $\pm$ 1.26% ADF, and 33.7% $\pm$ 1.91% NDF. Particle size distribution (% of DM) = 36.9% $\pm$ 1.9% long (>19mm), 41.4% $\pm$ 1.5% medium (8 to 19 mm), 12.1% $\pm$ 0.5% short (4 to 8 mm), and 9.6% $\pm$ 0.8% fine (<4 mm) particles.

⁵Alfalfa haylage had a DM of 40.1% and chemical composition (DM basis) of 17.6% $\pm$ 0.34% CP, 1.6% $\pm$ 0.09% starch, 35.4% $\pm$ 2.21% ADF, and 42.3% $\pm$ 3.67% NDF. Particle size distribution (% of DM) = 39.4% $\pm$ 1.8% long (>19mm), 42.5% $\pm$ 1.3% medium (8 to 19 mm), 9.7% $\pm$ 0.3% short (4 to 8 mm), and 8.4% $\pm$ 0.5% fine (<4 mm) particles.

⁶Supplied by Floradale Feed Mill Ltd. (Floradale, ON, Canada) including (DM basis) 23.7% soy hulls, 20.8% wheat shorts, 13.3% high bypass soybean meal, 13.1% beet pulp, 9.8% soybean meal (47%), 5.6% limestone calcium carbonate, 4.9% canola, 2.8% monocalcium phosphate, 2.4% molasses, 1.1% magnesium oxide, 1.0% fine salt, 0.57% granular sulfur, 0.5% organic mineral premix, 0.19% vitamin E (200KIU/kg), 0.15% vitamin ADE, 0.08% Availa ZN (Zinpro, Eden Prairie, MN), 0.04% Rumensin/Coban (Elanco, Greenfield, IN), and 0.01% Rovimix Biotin 20000 (DSM, Maastricht, the Netherlands).

Dry cow supplement had a DM of 90.24% and chemical composition (DM basis) of 19.9 $\pm$ 0.40% CP, 6.7 $\pm$ 0.03% starch, 22.1 $\pm$ 0.60% ADF, and 35.8 $\pm$ 0.88% NDF.

⁷Fat-microencapsulated anionic salt supplemented at 0.3 kg/cow/d DM, providing -2544mEq/d.

⁸DCAD calculated using the following equation: mEq/kg of DM = [(% Na $\times$ 435) + (% K $\times$ 256)] - [(% Cl $\times$ 282) + (% S $\times$ 624)].

by the difference between DM offered (calculated as the % of the diet made up by each fraction multiplied by the amount offered, in kg/d DM) and refused (calculated as the % of the refusals made up of each fraction multiplied by the amount of DM, in kg/d DM) from PSPS results of TMR. For each fraction, predicted consumption was the product of the total DMI of the TMR and DM of each

individual fraction from the fresh TMR. Sorting values equaling 100% indicated no sorting for that particle fraction occurred, values <100% represented sorting against the fraction in question, and a value >100% meant cows sorted in favor of that particular fraction.

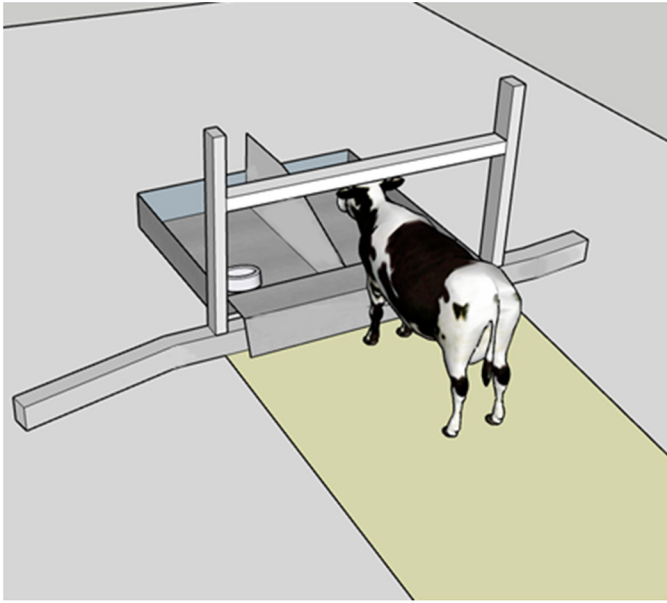


Figure 1. Diagram of the experimental setup consisting of a cow in a stall with a fitted split-bunk feed area.

Rumination and Urine pH

An electronic monitoring system (HR-TAG-LD, Merck & Co. Inc., Rahway, NJ) was used to monitor rumination activity, as validated by Merenda et al. (2020). As part of standard operating procedures at the research barn, all cows are continuously equipped with these rumination data loggers attached to their neck collars. Data were stored in 2-h intervals and transmitted wirelessly to a central processing unit. Files were downloaded at the end of the study and combined into a continuous record for each cow. Time spent ruminating was summarized daily for each cow (min/d).

Urine pH was measured 2×/wk for all cows during the adaptation and acceptability periods, and on d 1 and 3 of the preference test at 1000 h. Perineal stimulation was performed, and mid-stream urine samples were collected into a 20-mL plastic collection vial to measure pH with a handheld meter immediately following collection (Thermo Fisher Scientific, Waltham, MA). The pH meter was calibrated daily using buffers of pH 4, 7, and 10 (Thermo Fisher Scientific, Waltham, MA).

Personality Assessment

Cows were assessed for personality traits less than 2 wk following the completion of the preference test, between 1000 and 1200 h, using a combined arena test protocol adapted from Brasier et al. (2025). All cows had returned to their freestall housing for a minimum of 3 d before being involved in their personality assessment.

This test was conducted in an observational arena of 5 × 8 m, with metal fencing on all sides and rubber mats for flooring. In this arena, cows had limited visual contact with conspecifics and had access to a water trough. The center of the arena floor was marked with chalk with 2 square boxes, representing 1 m and 2 m from the center. The arena floor was also marked into quadrants using chalk. Cows were moved from their dry pens to the test arena by a familiar person. The combined arena test started when the cow entered the arena, and the gate was closed. The test consisted of 3 consecutive parts: (1) a novel arena test, in which the cow spent 10 min alone in the arena, followed by (2) a novel object test, in which an unknown object (a blue cooler with a lid: 60 × 60 × 60 cm) was placed in the center of the area while the cow was still in the arena and was removed after 10 min, directly followed by (3) a novel human test, in which an unknown human (3 different humans were used throughout: 2 females, 1 male) in standardized clothing (bright blue overalls with high visibility banding, which were unknown to the cows and not used by the barn staff), entered and stood at the center of the arena for 10 min. The novel human was instructed to face in the same direction for all tests, regardless of the location of the cow or what she was doing. After the test was completed, the familiar researcher led the cow back to her home pen. The arena floor was scraped clean between tests. Each 30 min arena test session was video recorded using 2 cameras to permit all corners of the pen to be viewed. One camera (MAX, GoPro Inc., San Mateo, CA) was attached to a metal bar just behind and above an arena wall to record all areas of the arena lengthwise. The other camera (Handycam Camcorder, Sony Corporation, Tokyo, Japan) was placed on a tripod of similar height behind the far corner of the adjacent wall to record the arena as a back-up. These video recordings were used to determine frequencies, durations, and latencies of behaviors displayed by the cows during each stage of the combined arena test (novel arena, novel object, and novel human; for a detailed ethogram see Table 3). These behaviors were chosen based on the studies of Müller and Schrader (2005), Rousing et al. (2006), Gibbons et al. (2009), Hedlund and Løvlie (2015), and Foris et al. (2018). All videos were coded by the same person after establishing interobserver reliability of $\kappa > 0.70$ for each test between 2 observers, and intra-observer reliability of $\kappa > 0.89$. In any cases where one camera did not record (through either malfunction or human error) and the cow was not visible on any video recording during the assessment, the time the cow was not visible was excluded from analyses. All time duration variables were converted to a percentage of test time, to account for time spent distracted or not in view, calculated as time spent performing the behavior, divided by the total test time. If applicable, the time the cow was

Table 2. Particle size distribution (mean $\pm$ SD) and nutrient content (mean $\pm$ SD) by particle size of the base dry cow diet and treatment diets

Item	Base diet	Treatment	
		ANIC ¹	PLCB ²
% DM retained on screen ³			
Long	4.3 $\pm$ 0.48	3.6 $\pm$ 0.26	3.9 $\pm$ 0.31
Medium	56.5 $\pm$ 1.31	53.4 $\pm$ 0.67	54.8 $\pm$ 0.61
Short	18.0 $\pm$ 0.79	18.9 $\pm$ 0.45	17.9 $\pm$ 0.37
Fine	21.2 $\pm$ 0.94	24.1 $\pm$ 0.51	23.4 $\pm$ 0.63
ADF ⁴ (% of screen DM)			
Long	32.5 $\pm$ 0.81	32.9 $\pm$ 1.21	33.5 $\pm$ 1.51
Medium	31.7 $\pm$ 0.43	31.7 $\pm$ 0.36	31.7 $\pm$ 0.89
Short	34.2 $\pm$ 0.16	33.1 $\pm$ 1.05	33.3 $\pm$ 0.63
Fine	31.0 $\pm$ 0.34	28.7 $\pm$ 0.74	31.6 $\pm$ 0.83
NDF ⁴ (% of screen DM)			
Long	48.0 $\pm$ 0.81	47.5 $\pm$ 1.19	48.3 $\pm$ 2.00
Medium	47.2 $\pm$ 0.87	46.3 $\pm$ 0.52	48.2 $\pm$ 1.38
Short	52.2 $\pm$ 0.33	49.1 $\pm$ 1.22	50.5 $\pm$ 0.55
Fine	45.9 $\pm$ 0.94	42.9 $\pm$ 0.89	46.1 $\pm$ 0.61
CP ⁴ (% of screen DM)			
Long	12.7 $\pm$ 0.34	13.3 $\pm$ 0.14	12.6 $\pm$ 0.28
Medium	12.7 $\pm$ 0.32	12.8 $\pm$ 0.21	12.4 $\pm$ 0.33
Short	10.6 $\pm$ 0.34	11.1 $\pm$ 0.62	9.9 $\pm$ 0.64
Fine	9.9 $\pm$ 0.45	11.8 $\pm$ 0.61	9.8 $\pm$ 0.28
Starch ⁴ (% of screen DM)			
Long	10.6 $\pm$ 0.93	11.1 $\pm$ 0.94	10.2 $\pm$ 1.19
Medium	12.7 $\pm$ 0.53	12.8 $\pm$ 0.43	11.6 $\pm$ 0.59
Short	14.7 $\pm$ 0.72	15.5 $\pm$ 1.26	15.7 $\pm$ 1.12
Fine	20.4 $\pm$ 0.48	20.1 $\pm$ 0.95	19.6 $\pm$ 0.91
Fat ⁴ (% of screen DM)			
Long	2.48 $\pm$ 0.11	2.56 $\pm$ 0.09	2.31 $\pm$ 0.22
Medium	2.31 $\pm$ 0.17	2.34 $\pm$ 0.10	2.62 $\pm$ 0.08
Short	2.45 $\pm$ 0.11	2.86 $\pm$ 0.14	4.14 $\pm$ 0.91
Fine	2.52 $\pm$ 0.10	6.99 $\pm$ 0.20	7.58 $\pm$ 0.30
Ash ⁴ (% of screen DM)			
Long	6.92 $\pm$ 0.34	6.79 $\pm$ 0.40	6.83 $\pm$ 0.23
Medium	7.23 $\pm$ 0.24	7.33 $\pm$ 0.47	7.48 $\pm$ 0.34
Short	5.88 $\pm$ 0.28	6.34 $\pm$ 0.29	5.99 $\pm$ 0.49
Fine	6.22 $\pm$ 0.39	7.14 $\pm$ 0.27	6.43 $\pm$ 0.17
DCAD ^{4,5} (mEq/kg DM)			
Long	170.6 $\pm$ 9.74	54.8 $\pm$ 17.8	142.0 $\pm$ 13.2
Medium	153.5 $\pm$ 8.66	62.7 $\pm$ 16.0	130.8 $\pm$ 9.75
Short	113.2 $\pm$ 14.6	2.38 $\pm$ 17.8	90.7 $\pm$ 2.83
Fine	82.0 $\pm$ 7.69	-242.5 $\pm$ 52.8	63.3 $\pm$ 15.6

¹Base diet with added fat-microencapsulated anionic salt (Jefo Nutrition Inc., Saint-Hyacinthe, QC, Canada) at 0.30 kg/d DM per cow (formulated DCAD = -101.5 mEq/kg DM; 49% fat).

²Base diet with added fat placebo (Jefo Nutrition Inc., Saint-Hyacinthe, QC, Canada) at 0.15 kg/d DM per cow (formulated DCAD = +64 mEq/kg DM; 98% fat).

³Particle size was determined by a Penn State Particle Separator (Heinrichs, 2013) utilizing a 19-mm screen (long), 8-mm screen (medium), 4-mm screen (short), and a pan (fine).

⁴Values were taken from chemical analysis of TMR samples separated by Penn State Particle Separator by particle size.

⁵DCAD calculated using the following equation: mEq/kg of DM = [(% Na $\times$ 435) + (% K $\times$ 256)] - [(% Cl $\times$ 282) + (% S $\times$ 624)].

not visible during the assessment was subtracted from the total assessment time before calculation.

Statistical Analysis

All statistical analyses were carried out using SAS 9.4 (SAS Institute Inc., Cary, NC). Statistical significance was declared at $P \leq 0.05$ and tendencies were declared as $0.05 < P \leq 0.10$. Before analyses, data were screened for normality by visually assessing residual data plots and

Table 3. Ethogram used for scoring personality behavior videos¹

Behavior	Definition
Recorded for novel arena, object and human test	
Vocalizations (count) ²	All types of vocalizations made by the cow
Vigilance (s)	Cow raises their head above the withers while stationary
Attraction (s)	Cow raises their head above the withers while stationary looking in the direction of the home pen
Attention (s)	Cow focuses their attention to the object or human
Locomotion (s)	Cow moves all 4 legs to cover distance
Exploratory (s)	Cow sniffs or licks an object or surface in the testing arena, excluding the novel object or human
Escape (s)	Cow head is through or over the railing of the testing arena
Changing quadrants (count)	When a cow switches to a new quadrant
Recorded for novel object and human test only	
Time spent at a certain distance (>2m away, within 1m, and contact) from the novel object/human (s)	Any part of the cow's body in physical contact with the novel object or person
Latency to contact novel object/human (s)	Time from the start of the test to first contact with the novel object or human
Time spent behind novel person (s)	Time spent in the 2 quadrants behind the novel person's back

¹From Brasier et al. (2023). Originally adapted from Müller and Schrader (2005), Rousing et al. (2006), Gibbons et al. (2009), Hedlund and Løvlie (2015), and Foris et al. (2018).

²Separate events were all at least 10 s between each count.

using the Shapiro-Wilk test in the UNIVARIATE procedure in SAS. The assumptions for normality were met for all variables in the acceptability phase and the preference test. For the personality assessment, the behaviors of vocalizations and time spent exploring, attraction to the herd during the novel object and human test, and time spent walking and time spent behind the novel human in the novel human test were square root transformed to achieve normality. Behaviors of quadrant switches, vigilance, time spent escaping, latency to contact, time in contact, time spent <1 m, and attention toward the novel object and human, attraction to the herd in the arena test, and time spent walking in the arena and novel object test were all log10 transformed to achieve normality.

For personality assessment, the behaviors scored from different tests within the combined arena test were analyzed for association using the SAS CORR procedure. If the behaviors were significantly associated, defined as $P < 0.05$, then the results were averaged across all the tests in the combined arena test that measured the specific behavior, which for this study was only the behavior of quadrant changes.

Principal components analysis (PCA) was performed to identify personality traits using the FACTOR procedure of SAS with varimax rotation by condensing associated behavioral measures from the combined arena test. Criteria for PCA of animal behavior data followed Budaev (2010), requiring a sampling adequacy of >0.50 (test of validity related to subject: variable ratio). After examination of scree plots, factors with eigenvalues greater than 1.0 were retained, and variables with communality estimates <0.30 were removed; this resulted in 2 factors. Behaviors retained in the PCA were latency to contact the novel human, time in contact with the

novel human, walking in the arena and object tests, and vigilance in the object test. High loadings of behaviors from the tests on each factor were considered equal to or greater than ± 0.63 . For each factor, the score for each cow was extracted, indicating where each cow lies along an axis from highly negative to highly positive for that specific factor.

The data from the acceptability phase and preference test were analyzed in separate mixed-effect linear regression models using the MIXED procedure of SAS. For the acceptability phase, DMI, feed sorting, rumination, and urine pH were analyzed in repeated measures, mixed-effect linear regression models. These models included the fixed effects of treatment, period, order, day, and the interaction between treatment and day. The subject of the repeated statement was cow within cohort. Cohort was included as a random effect. Compound symmetry was the selected covariance structure, as determined by the Schwarz's Bayesian information criterion. Base diet DMI was included as a covariate in the model for DMI. When an interaction of treatment and day was detected ($P \leq 0.1$), treatment means were compared within day. Factor score, from the above-described PCA, as well as the factor score and treatment interaction, were tested as other potential predictors in these models, but neither were retained ($P > 0.1$). Similarly, the side of the bunk that each diet was presented on, and side of the bunk and treatment interaction were tested as other potential predictors in these models, but neither were retained ($P > 0.1$).

For the preference test, the proportion of daily intake for each diet consumed, feed sorting, rumination, and urine pH were again analyzed in repeated measures, mixed-effect linear regression models. All models in-

Table 4. Effect of dry cow dietary treatment on intake, feed sorting, rumination, and urine pH during the baseline and treatment period

Item ²	Treatment ¹			SEM ³	P-value		
	Base ²	ANIC	PLCB		Treatment	Day	Treatment × day
DMI (kg/d) ⁴⁺	19.9	19.2	19.5	0.38	0.37	0.12	0.54
Feed sorting ^{5,6} (%)							
Long	100.7	100.7	97.2*	1.23	0.03	0.03	0.36
Medium	102.2*	102.2*	101.8*	0.33	0.28	0.09	0.14
Short	99.4*	99.4*	99.1*	0.20	0.26	0.74	0.24
Fine	95.1*	95.2*	96.5*	0.74	0.11	0.008	0.02
Rumination ⁷ (min/d)	546.2	521.3	532.4	9.56	0.41	0.001	0.47
Urine pH ⁸	8.2	7.6	8.2	0.05	<0.001	0.04	0.86

¹Treatments of either ANIC (base diet with added fat-microencapsulated anionic salt at 0.30 kg/d DM per cow (formulated DCAD = −101.5 mEq/kg of expected DMI; 49% fat) or PLCB (base diet with added fat placebo at 0.15 kg/d DM per cow (formulated DCAD = +64 mEq/kg of expected DMI; 98% fat).

²Baseline averages, collected from d −49 to −42, were included as possible covariates and retained when $P < 0.05$ (indicated with a +).

³Highest SEM across treatments is presented.

⁴Data were collected for 7 d during the baseline period and over 7 d for 24 cows on each treatment.

⁵Particle size was determined by a Penn State Particle Separator (Heinrichs, 2013) utilizing a 19-mm screen (long), 8-mm screen (medium), 4-mm screen (short), and a pan (fine). A sorting % = 100 means no sorting occurred, sorting % >100 means sorting occurred in favor of that particle fraction, and sorting % <100 means sorting occurred against that particle fraction.

⁶Data were collected for 24 cows for 3 d during the baseline period, and 3 d on each treatment.

⁷Data were collected for 24 cows for 7 d during the baseline period and over 7 d on each treatment.

⁸Data were collected for 24 cows for 2 d during the baseline period, and 2 d on each treatment.

* $P < 0.05$ = difference in sorting from 100%.

cluded the fixed effects of treatment and day. The subject of the repeated statement was cow within cohort. Cohort was included as a random effect. Compound symmetry was the selected covariance structure as determined by the Schwarz's Bayesian information criterion. Factor score, from the above-described PCA, as well as the factor score and treatment interaction were tested as potential predictors in these models. Factor score was retained ($P < 0.1$), but no factor score and treatment interactions were detected ($P > 0.1$).

RESULTS

Acceptability Phase

During the acceptability phase, no treatment differences were detected for daily DMI or rumination time (Table 4). A treatment difference in urine pH was observed, with cows having lower urine pH when consuming the ANIC diet compared with when on the PLCB diet (Table 4). A treatment difference was detected in the sorting of long (>19 mm) ration particles, whereby cows sorted against the long particles if consuming the PLCB diet but did not sort for or against those particles on the ANIC diet (Table 4). No treatment differences were detected for the sorting of medium (<19 and >8 mm) or short (<8 and >4 mm) ration particles, with cows sorting for the medium fraction, and against the short fraction, on both treat-

ments (Table 4). On both treatments, cows consumed less fine (<4 mm) ration particles than expected (Table 4). A treatment by day interaction for the sorting of these fine ration particles was detected; however, no individual comparisons of treatment by day were different ($P > 0.1$).

Preference Test

During the 3-d preference test, there was no difference detected in the proportion (% of total DMI) of the ANIC and PLCB diets consumed (Table 5). Within the PLCB diet, cows sorted against the long ration particles, with a treatment difference from the ANIC diet (Table 5). Cows tended to sort the short ration particles differently between treatments (Table 5). No treatment differences were detected for the sorting behavior of the medium and fine particles, with cows sorting for the medium particles and against the fine particles within both diets (Table 5). No treatment differences were detected for daily rumination time or urine pH during the preference phase (Table 5).

Personality Traits and Associations with Behavior

The loadings for each factor extracted from the PCA are reported in Table 6. The PCA had a sampling adequacy of 0.54. Factor 1 explained 41.5% of the total variance and contained high positive loadings for walking during

Table 5. Effect of dry cow dietary treatment on proportion of intake, feed sorting, rumination, and urine pH during the 3-d preference test

Variable	Treatment ¹		SEM ²	P-value
	ANIC	PLCB		
Preference ratio, % of total DMI ³	48.6	51.4	1.73	0.26
Feed sorting (%) ^{4,5}				
Long	95.9	82.5*	7.37	0.05
Medium	105.4*	107.3*	1.24	0.22
Short	101.1	98.8	2.12	0.07
Fine	87.2*	85.8*	2.32	0.66
Rumination (min/d) ⁶	500.4	499.3	50.3	0.99
Urine pH ⁷	7.9	8.1	0.31	0.59

¹Treatments of either ANIC (Base diet with added fat-microencapsulated anionic salt at 0.30 kg/d DM per cow (formulated dietary DCAD = −101.5 mEq/kg of expected DMI; 49% fat) or PLCB (Base diet with added fat placebo at 0.15 kg/d DM per cow (formulated dietary DCAD = +64 mEq/kg of expected DMI; 98% fat).

²Standard error of the highest treatment group.

³DMI of both ANIC and PLCB diets were measured each day and the preference ratio was calculated by dividing the DMI of each diet by total DMI during the 3 d preference test for 24 cows.

⁴Particle size was determined by a Penn State Particle Separator (Heinrichs, 2013) utilizing a 19-mm screen (long), 8-mm screen (medium), 4-mm screen (short), and a pan (fine). A sorting % = 100 means no sorting occurred, sorting % > 100 means sorting occurred in favor of that particle fraction, and sorting % < 100 means sorting occurred against that particle fraction.

⁵Data were collected for 24 cows for 1 d during the preference test.

⁶Data were collected for 24 cows each day during the preference test.

⁷Data were collected for 24 cows for 2 d during the preference test.

* $P < 0.05$ = difference in sorting from 100%.

the arena and object tests, and vigilance during the object test. Cows who scored high on factor 1 were interpreted as being “active.” Factor 2 explained 26.9% of the total variance and contained high positive loadings for time in contact with the novel human, and high negative loadings for latency to contact the novel human. Cows who scored high on factor 2 were interpreted as “bold.”

Cows that scored low for the PCA trait, bold, consumed a higher portion of the ANIC diet (% of total

DMI) during the preference test than the more bold cows ($P = 0.04$; Figure 2). Cows that scored low for PCA trait, active, sorted more for the medium particles of both the ANIC and PLCB diets during the preference test ($P = 0.02$; Figure 3a). More active cows sorted more for the short particles of both diets ($P = 0.01$; Figure 3b). Less active cows tended to sort more against fine particles during the preference test to a greater degree than their active counterparts ($P = 0.07$; Figure 3c).

Table 6. Coefficients (loadings) of the eigenvectors for the factors extracted from each principal components analysis of behaviors recorded when cows ($n = 24$) were subjected to a combined arena test^{1,2}

Test	Behavior ³	Factor 1	Factor 2
Combined arena	Latency to contact the novel human	0.243	−0.814*
	Time in contact with the novel human	0.129	0.856*
	Walking during the arena test	0.765*	−0.005
	Walking during the object test	0.786*	−0.058
	Vigilance during the object test	0.860*	−0.068
Eigenvalues		2.08	1.35
Variance explained (%)		41.5	26.9
Interpretation ⁴		Active	Bold

¹Combined arena test consisted of successive exposure to a novel arena (10 min), novel object (10 min), and novel human (10 min).

²High loadings ($\geq |0.63|$) are shown with an asterisk, indicating behavior variables that were highly associated within each factor.

³Variables were $\log_{10}$ transformed to meet assumption of normality.

⁴Personality traits were interpreted based on behaviors exhibited during the combined arena test that loaded highly (either positively or negatively) onto each factor.

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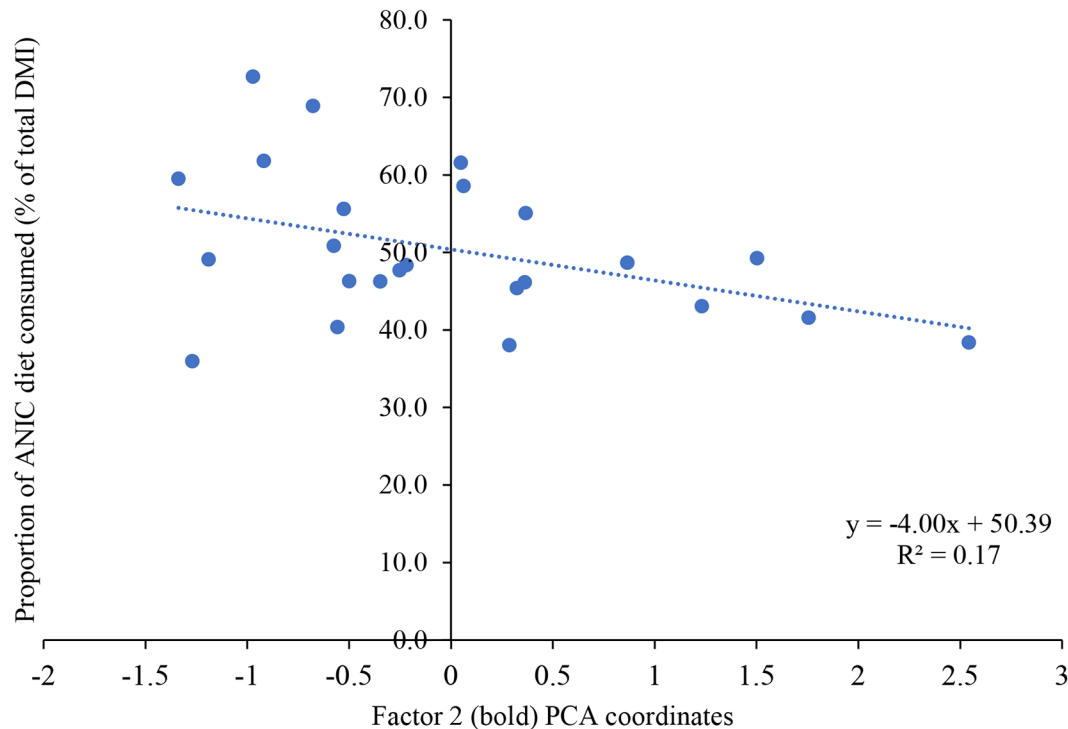


Figure 2. Association between principal component analysis trait, bold, of the personality test with the proportion of ANIC (base diet with added fat-microencapsulated anionic salt at 0.30 kg/d DM (formulated DCAD = -101.5 mEq/kg DM; 49% fat) diet consumed while being offered PLCB (base diet with added fat placebo at 0.15 kg/d DM (formulated dietary DCAD = $+64$ mEq/kg DM; 98% fat) diet simultaneously during the 3-d preference test (n = 24 cows)

DISCUSSION

Acceptability and Preference of Treatment Diets

The goal of the study was to determine the acceptability and preference of nonlactating dairy cows for a fat-microencapsulated anionic salt, and to determine if individual cow personality affected these responses. As hypothesized, in the acceptability phase, there was no difference in DMI between treatment diets, despite the intended result of cows consuming the ANIC diet having a reduced urine pH. Additionally, during the preference test phase, there were no differences between the proportion of diets consumed. Hultquist and Casper (2016) similarly demonstrated a lack of palatability issues when encapsulating potassium carbonate with free fatty acids. Microencapsulation is a technique used to enhance the delivery and effectiveness of feed additives in dairy cows (Morey et al., 2011; Vittorazzi et al., 2022). This process involves coating feed ingredients with a protective layer, which can improve the stability, palatability, and targeted release of nutrients or additives in the digestive system of cows (Morey et al., 2011; Vittorazzi et al., 2022). Several researchers have investigated the effects of encapsulated feed additives on rumen-degradation, digestibility, and production (Kim et al., 2020; Zhang et al., 2023; Lago

et al., 2024), but have not investigated palatability. A review by Coupland and Hayes (2014) discussed physical encapsulation as a means to mask bitter taste in human pharmaceuticals. Medium chain triglycerides have been demonstrated to improve the palatability of oral pharmaceutical liquids in humans (Bahal et al., 2003), but no research appears to have been done on fat-specific microencapsulation of feed additives to improve palatability.

No differences in daily rumination time were observed between treatments during the acceptability phase. Given that DMI and rumination time are positively associated in dairy cows (Johnston and DeVries, 2018), the lack of DMI difference in the acceptability phase explains the similarity in rumination time. This is the first study, to our knowledge, that tested a fat-microencapsulated anionic salt against a fat placebo, so this result is an important piece in determining whether fat-microencapsulation can mitigate any potential palatability problems caused by the feeding of anionic salts to dairy cows (Oetzel, 1988; Goff et al., 2004).

Interestingly, there was a numerical difference in average rumination time between the acceptability phase and the preference test. Rumination time was lesser during the preference test, likely due to increased feeding activ-

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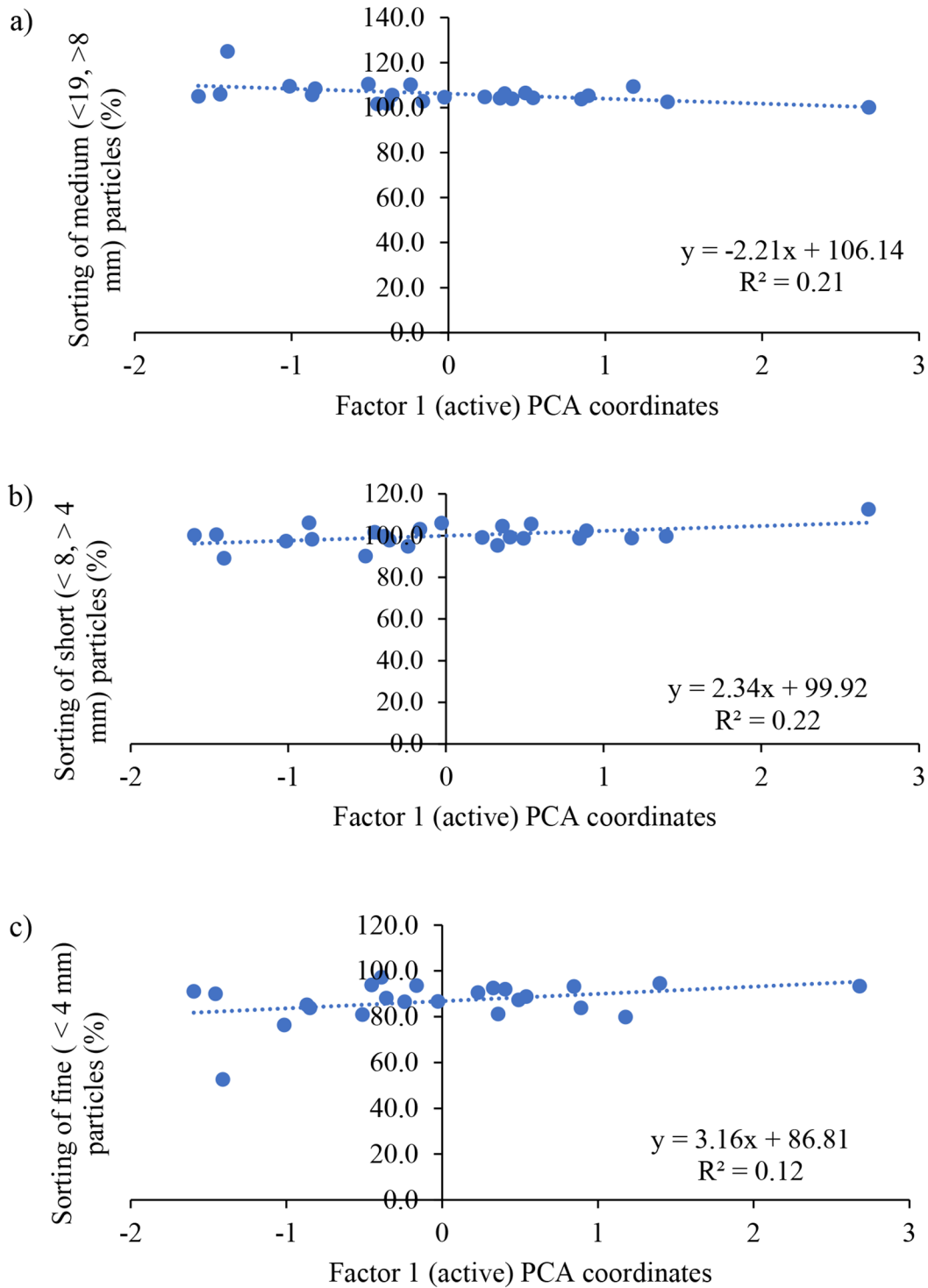


Figure 3. Association between principal component analysis trait, active, and sorting (%) of (a) medium (<19, >8 mm) particles, (b) short (<8, >4 mm) particles, and (c) fine (<4 mm) particles during the 3-d preference test where cows had access to both ANIC (base diet with added fat-microencapsulated anionic salt at 0.30 kg/d DM (formulated DCAD = -101.5 mEq/kg DM; 49% fat) and PLCB (base diet with added fat placebo at 0.15 kg/d DM (formulated DCAD = +64 mEq/kg DM; 98% fat) diets, with sorting values averaged between both diets (n = 24 cows).

ity and greater total DM offered, as the cows were presented with both of the diets simultaneously, which may have altered the normal feeding and rumination patterns (Schirmann et al., 2012) as compared with the single-diet acceptability phase.

There is no widely accepted DCAD value that should be fed to cows to maximize the benefits of improved Ca metabolism postpartum, while retaining proper DMI prepartum, but values in the literature typically range from -200 mEq/kg DM to -100 mEq/kg DM (Santos et al., 2019; Caixeta et al., 2020). High metabolic acidification is one of the reasons thought to limit DMI when feeding acidogenic diets (Vagnoni and Oetzel, 1998; Santos et al., 2019). The level of acidification in the ANIC diet (DCAD = -22.4 mEq/kg of DM consumed) would not have been expected to have any effect on the DMI or acceptability of the diet on the basis of the level of metabolic acidosis (Vagnoni and Oetzel, 1998; Santos et al., 2019).

Reduced urine pH of the cows while on the ANIC diet, as demonstrated, was expected due to the induced metabolic acidosis caused by feeding anionic salts. Although this was achieved, the urine pH values of all cows on both treatments were slightly higher than expected. Some possible reasons for the unexpectedly high urinary pH values, across cows, include the elevated potassium levels in the ANIC and PLCB diets, at 1.24% and 1.29% DM, respectively, which are on the upper end of what is typically fed to far-off dry cows. Higher dietary potassium can reduce the effectiveness of acidification and contribute to increased urinary pH. Additionally, cows had a much higher DMI than expected, consuming 19.2 kg/d DM and 19.5 kg/d DM for the ANIC and PLCB diets, respectively. The diets were formulated for cows with a DMI level of 15 kg/d DM, so this increased intake may have diluted the effects of the anionic salt, which was supplemented on a fixed amount basis of 0.30 kg/d DM per cow, which was not adjusted throughout the trial. Another possible explanation for the higher than expected urine pH values is the sorting behavior exhibited during the acceptability phase. Cows sorted against both the short and fine fractions of both treatment diets, which is where the fat-microencapsulated anionic salt and fat placebo appeared to primarily be located, based on fat and mineral analyses of the fractions (Table 2). Across diets, lower DCAD values were found in those short and fine fractions; thus, more sorting against these particle fractions may have caused their urine pH to be higher than expected. Various researchers have reported urine pH values alongside DCAD values (Lopera et al., 2018; Santos et al., 2019; Melendez and Chelikani, 2022). In a meta-analysis by Santos et al. (2019), they demonstrated that for a DCAD value of -30 mEq/kg DM, cows had a

urine pH around 6.8. Based on this, for the DCAD level in the ANIC diet in the present study, urine pH values between 6.5 and 7 would have been expected, as opposed to the mean value of 7.6 in the present study. Similarly, the urine pH of 8.2 for cows consuming the PLCB diet (DCAD = 168.7 mEq/kg of DM consumed) was slightly higher than expected. With a DCAD value of 152.9 mEq/kg DM, pH values between 7.5 and 8 would be expected (Santos et al., 2019). Given this higher than expected urine pH of cows on both treatments, it is likely that this was driven the above-described limitations, including the high dietary potassium, high overall DMI, and sorting against the short and fine particle fractions of the diets.

Personality Traits and Associations with Behavior

The PCA identified 2 common personality traits, “bold” and “active,” unlike in other studies where 3 or more have been typically identified (Neave et al., 2019; Schwanke et al., 2022). Brasier et al. (2025) also identified only 2 personality traits, in a study on the effect of training method and dairy cow personality traits on adaptation to an automated milking system, although they combined the traits of “explore” and “active,” while traits were not combined in the present study. The “explore” trait was not included in our analysis, as it was not considered normally distributed due to a limited spread of factor scores from the PCA.

Cows that scored lower for boldness consumed a higher portion of the ANIC diet (% of total DMI) during the preference test than the more bold cows, while the more bold cows were more likely to equally consume the feeds. The first result differs from the literature, where bolder animals are more explorative at the feed bunk, and are more willing to sample novel feeds (Neave et al., 2018). For example, Bergvall et al. (2011) demonstrated that domestic deer that spent more time in close proximity to, or investigating a novel object spent more time investigating novel feeds. Similarly, beef heifers that spent more time investigating a novel object were more likely to be positioned near the front of the herd when grazing, positioning themselves to sample feeds before others (Ramseyer et al., 2009). We are unsure why the less bold cows preferred the ANIC diet more than the PLCB diet, as there were no detected effects of side of diet during the statistical analysis. One possible explanation is that the less bold cows consumed more of whichever diet they consumed first, as they may have been less willing to sample feeds, whereas the more bold cows were more likely to consume both feeds. This theory fits with research conducted by Bergvall et al. (2011) and Ramseyer et al. (2009), but more research is needed to confirm this.

We also detected associations between personality traits and the degree of feed sorting during the preference test. Although on average, cows sorted for medium particles in both ANIC and PLCB diets, the less active cows sorted for these particles to a higher degree. Interestingly, the opposite was observed for the sorting of short and fine particles, with less active cows sorting against the short and fine particles to a higher degree than their more active counterparts. Cattle with increased activity are less feed efficient than those with a lower energy expenditure (Llonch et al., 2018). Greter and DeVries (2011) demonstrated a relationship between the degree of feed sorting and DMI, with cows who sort less having a higher DMI. In the present study, active cows sorted feed to a greater degree than their less active counterparts, and are likely using more time and energy on being active and sorting, as opposed to simply consuming feed. This can lead to potential challenges with poor DMI, which can have negative consequences for maintaining health and production (Greter and DeVries, 2011). Additionally, animals that expend more energy through increased activity may need to consume more of the higher energy components in the diet that are typically distributed within the smaller particles. Moore and DeVries (2020) demonstrated that cows increase their sorting for short and fine particles when experiencing a diet-induced negative energy balance. If the more active cows had a higher energy expenditure and, therefore, required more energy for maintenance, this may explain why less active cows chose to consume more medium particles and fewer fine particles than cows who scored higher for the active trait. These associations between personality traits and feed preference behaviors, though exploratory, offer a novel perspective on explaining cow-level variation, which could be relevant for future precision nutrition strategies.

Study Limitations

There are a few limitations regarding this study that may have contributed to the observed results. As discussed above, the level of acidification achieved, on a per kilogram DM basis, was lower than originally targeted (actual DCAD: -22 mEq/kg DM vs. formulated: -101 mEq/kg DM). Although the cows still consumed the targeted $-2,544$ mEq/d from the supplemental anionic salt, the reduced dietary acidification concentration may have influenced the outcomes observed in the study. Additionally, the cows included in this study were recently dried off and were in the far-off dry period. Negative DCAD diets are typically fed to close-up dry cows, which typically have a lower DMI. This may have caused the dilution of the anionic salt observed with the larger than expected DMI consumed throughout the study. The

length of periods in this study (1 wk for each part of the crossover during the acceptability phase) may have been too short for cows to fully form associations between the 2 treatment diets. Longer periods may have altered the degree of acceptability and preference for either diet. Another limiting factor in this study may be the design of the split-bunk bins used during both the acceptability and preference phases. Unlike a traditional feed bunk, the experimental bins had surrounding walls, which may have limited the sorting ability of the cows. Cows typically sort by using their prehensile tongues to sweep the feed and their noses to push less desirable portions of the TMR away (Fish and DeVries, 2012; Miller-Cushon and DeVries, 2017), and the walls on the bins and frequent feed push-ups may have inhibited their sorting behavior. Future studies looking at feed sorting between different diets should consider adapting a divider to fit directly onto the feed bunk without exterior walls. Finally, although personality is generally considered to remain stable across contexts, including under moderate stress associated with management changes (Brasier et al., 2023; Schwanke et al., 2024b), we recognize that full behavioral expression—particularly of traits such as exploration and activity—can be moderated by space availability and housing conditions, as demonstrated by Russell et al. (2023). Thus, it is possible that the behaviors exhibited during the personality test may have been affected by the limited time that cows spent in their regular freestall housing following their time in the tiestall environment.

CONCLUSIONS

The results of this study indicate that dry cows showed no preference between a TMR containing a fat-microencapsulated anionic salt or a fat placebo, and demonstrated similar acceptability of both diets. Thus, using a fat-microencapsulated anionic salt in a close-up TMR may be effective in helping maintain DMI when using a negative DCAD hypocalcemia prevention strategy. In terms of personality, less bold cows consumed more of the TMR containing a fat-microencapsulated anionic salt than their bolder counterparts, and cows with both more bold and active traits were associated with differences in sorting behavior, demonstrating that certain personality traits may affect feed sorting behavior in dairy cows.

NOTES

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REFERENCES

- Bahal, S. M., J. M. Romansky, and F. J. Alvarez. 2003. medium chain triglycerides as vehicle for palatable oral liquids. *Pharm. Dev. Technol.* 8:111–115. <https://doi.org/10.1081/pdt-120017518>.
- Bergvall, U. A., A. Schäpers, P. Kjellander, and A. Weiss. 2011. Personality and foraging decisions in fallow deer, *Dama dama*. *Anim. Behav.* 81:101–112. <https://doi.org/10.1016/j.anbehav.2010.09.018>.
- Brasier, J. E., A. J. Schwanke, R. Bergeron, D. B. Haley, and T. J. DeVries. 2025. Effect of training method and dairy cow personality traits on adaptation to an automated milking system. *J. Dairy Sci.* 108:885–899. <https://doi.org/10.3168/jds.2024-25420>.
- Brasier, J. E., A. J. Schwanke, and T. J. DeVries. 2023. Effects of dairy cows' personality traits on their adaptation to an automated milking system following parturition. *J. Dairy Sci.* 106:7191–7202. <https://doi.org/10.3168/jds.2022-23176>.
- Budaev, S. V. 2010. Using principal components and factor analysis in animal behaviour research: Caveats and guidelines. *Ethology* 116:472–480. <https://doi.org/10.1111/j.1439-0310.2010.01758.x>.
- Caixeta, L. S., W. J. Weber, D. M. Johnson, J. Faser, B. M. Visser, and B. A. Crooker. 2020. Effects of anionic supplement source in prepartum negative dietary cation-anion difference diets on serum calcium, feed intake, and lactational performance of multiparous dairy cows. *J. Dairy Sci.* 103:4302–4314. <https://doi.org/10.3168/jds.2019-16991>.
- Charbonneau, E., D. Pellerin, and G. R. Oetzel. 2006. Impact of lowering dietary cation-anion difference in nonlactating dairy cows: A meta-analysis. *J. Dairy Sci.* 89:537–548. [https://doi.org/10.3168/jds.S0022-0302\(06\)72116-6](https://doi.org/10.3168/jds.S0022-0302(06)72116-6).
- Coupland, J. N., and J. E. Hayes. 2014. Physical approaches to masking bitter taste: Lessons from food and pharmaceuticals. *Pharm. Res.* 31:2921–2939. <https://doi.org/10.1007/s11095-014-1480-6>.
- Fish, J. A., and T. J. DeVries. 2012. Short communication: Varying dietary dry matter concentration through water addition: Effect on nutrient intake and sorting of dairy cows in late lactation. *J. Dairy Sci.* 95:850–855. <https://doi.org/10.3168/jds.2011-4509>.
- Foris, B., M. Zebunke, J. Langbein, and N. Melzer. 2018. Evaluating the temporal and situational consistency of personality traits in adult dairy cattle. *PLoS One* 13:e0204619. <https://doi.org/10.1371/journal.pone.0204619>.
- Gibbons, J., A. Lawrence, and M. Haskell. 2009. Responsiveness of dairy cows to human approach and novel stimuli. *Appl. Anim. Behav. Sci.* 116:163–173. <https://doi.org/10.1016/j.applanim.2008.08.009>.
- Goff, J. P., R. Ruiz, and R. L. Horst. 2004. Relative acidifying activity of anionic salts commonly used to prevent milk fever. *J. Dairy Sci.* 87:1245–1255. [https://doi.org/10.3168/jds.S0022-0302\(04\)73275-0](https://doi.org/10.3168/jds.S0022-0302(04)73275-0).
- Greter, A., and T. DeVries. 2011. Effect of feeding amount on the feeding and sorting behaviour of lactating dairy cattle. *Can. J. Anim. Sci.* 91:47–54. <https://doi.org/10.4141/CJAS10067>.
- Hedlund, L., and H. Løvlie. 2015. Personality and production: Nervous cows produce less milk. *J. Dairy Sci.* 98:5819–5828. <https://doi.org/10.3168/jds.2014-8667>.
- Hultquist, K. M., and D. P. Casper. 2016. Palatability evaluation of free fatty acid encapsulated potassium carbonate as a feed ingredient for lactating dairy cows fed a total mixed ration. *Prof. Anim. Sci.* 32:328–332. <https://doi.org/10.15232/pas.2015-01485>.
- Johnston, C., and T. J. DeVries. 2018. Short communication: Associations of feeding behavior and milk production in dairy cows. *J. Dairy Sci.* 101:3367–3373. <https://doi.org/10.3168/jds.2017-13743>.
- Kim, T.-B., J.-S. Lee, S.-Y. Cho, and H.-G. Lee. 2020. In vitro and in vivo studies of rumen-protected microencapsulated supplement comprising linseed oil, vitamin e, rosemary extract, and hydrogenated palm oil on rumen fermentation, physiological profile, milk yield, and milk composition in dairy cows. *Animals (Basel)* 10:1631. <https://doi.org/10.3390/ani10091631>.
- Lago, R. V. P., G. J. Wolschick, M. H. Signor, G. C. Giraldi, V. L. Molosse, G. L. Deolindo, B. G. O. Cecere, A. L. R. Brunetto, D. C. Cucco, P. D. B. Benedetti, B. C. Deon, R. Wagner, B. F. Bissacotti, J. Nauderer, C. T. K. Jung, G. V. Kozloski, M. M. Bajay, and A. S. Da Silva. 2024. A mixture of free and microencapsulated essential oils combined with turmeric and tannin in the diet of cattle in the growing and finishing phase: A new tool to enhance productivity. *Anim. Feed Sci. Technol.* 315:116033. <https://doi.org/10.1016/j.anifeeds.2024.116033>.
- Leonardi, C., and L. E. Armentano. 2003. Effect of quantity, quality, and length of alfalfa hay on selective consumption by dairy cows. *J. Dairy Sci.* 86:557–564. [https://doi.org/10.3168/jds.S0022-0302\(03\)73634-0](https://doi.org/10.3168/jds.S0022-0302(03)73634-0).
- Llonch, P., M. Somarriba, C. A. Duthie, S. Troy, R. Roehe, J. Rooke, M. J. Haskell, and S. P. Turner. 2018. Temperament and dominance relate to feeding behaviour and activity in beef cattle: Implications for performance and methane emissions. *Animal* 12:2639–2648. <https://doi.org/10.1017/S1751731118000617>.
- Lopera, C., R. Zimpel, A. Vieira-Neto, F. R. Lopes, W. Ortiz, M. Poindexter, B. N. Faria, M. L. Gambarini, E. Block, C. D. Nelson, and J. E. P. Santos. 2018. Effects of level of dietary cation-anion difference and duration of prepartum feeding on performance and metabolism of dairy cows. *J. Dairy Sci.* 101:7907–7929. <https://doi.org/10.3168/jds.2018-14580>.
- McArt, J. A., and R. C. Neves. 2020. Association of transient, persistent, or delayed subclinical hypocalcemia with early lactation disease, removal, and milk yield in Holstein cows. *J. Dairy Sci.* 103:690–701. <https://doi.org/10.3168/jds.2019-17191>.
- Meagher, R. K., D. M. Weary, and M. A. G. von Keyserlingk. 2017. Some like it varied: Individual differences in preference for feed variety in dairy heifers. *Appl. Anim. Behav. Sci.* 195:8–14. <https://doi.org/10.1016/j.applanim.2017.06.006>.
- Melendez, P., and P. K. Chelikani. 2022. Review: Dietary cation-anion difference to prevent hypocalcemia with emphasis on over-acidification in prepartum dairy cows. *Animal* 16:100645. <https://doi.org/10.1016/j.animal.2022.100645>.
- Melin, M., H. Wiktorsson, and L. Norell. 2005. Analysis of feeding and drinking patterns of dairy cows in two cow traffic situations in automatic milking systems. *J. Dairy Sci.* 88:71–85. [https://doi.org/10.3168/jds.S0022-0302\(05\)72664-3](https://doi.org/10.3168/jds.S0022-0302(05)72664-3).
- Merenda, V. R., C. C. Figueiredo, T. D. González, and R. C. Chebel. 2020. Technical note: Validation of a system for monitoring individual behavior of Holstein cows. *J. Dairy Sci.* 103:7425–7430. <https://doi.org/10.3168/jds.2019-17922>.
- Miller-Cushon, E. K., and T. J. DeVries. 2017. Feed sorting in dairy cattle: Causes, consequences, and management. *J. Dairy Sci.* 100:4172–4183. <https://doi.org/10.3168/jds.2016-11983>.

- Moore, S. M., and T. J. DeVries. 2020. Effect of diet-induced negative energy balance on the feeding behavior of dairy cows. *J. Dairy Sci.* 103:7288–7301. <https://doi.org/10.3168/jds.2019-17705>.
- Morey, S. D., L. K. Mamedova, D. E. Anderson, C. K. Armendariz, E. C. Titgemeyer, and B. J. Bradford. 2011. Effects of encapsulated niacin on metabolism and production of periparturient dairy cows. *J. Dairy Sci.* 94:5090–5104. <https://doi.org/10.3168/jds.2011-4304>.
- Morris, T. 1999. *Experimental Design and Analysis in Animal Sciences*. CABI Publishing, New York, NY.
- Müller, R., and L. Schrader. 2005. Individual consistency of dairy cows' activity in their home pen. *J. Dairy Sci.* 88:171–175. [https://doi.org/10.3168/jds.S0022-0302\(05\)72675-8](https://doi.org/10.3168/jds.S0022-0302(05)72675-8).
- Neave, H. W., J. H. C. Costa, J. B. Benetton, D. M. Weary, and M. A. G. von Keyserlingk. 2019. Individual characteristics in early life relate to variability in weaning age, feeding behavior, and weight gain of dairy calves automatically weaned based on solid feed intake. *J. Dairy Sci.* 102:10250–10265. <https://doi.org/10.3168/jds.2019-16438>.
- Neave, H. W., D. M. Weary, and M. A. G. von Keyserlingk. 2018. Review: Individual variability in feeding behaviour of domesticated ruminants. *Animal* 12:s419–s430. <https://doi.org/10.1017/S1751731118001325>.
- Oetzel, G. R. 1988. Parturient paresis and hypocalcemia in ruminant livestock. *Vet. Clin. North Am. Food Anim. Pract.* 4:351–364. [https://doi.org/10.1016/s0749-0720\(15\)31053-7](https://doi.org/10.1016/s0749-0720(15)31053-7).
- Ramseyer, A., A. Boissy, B. Thierry, and B. Dumont. 2009. Individual and social determinants of spontaneous group movements in cattle and sheep. *Animal* 3:1319–1326. <https://doi.org/10.1017/S1751731109004790>.
- Rousing, T., J. H. Badsberg, I. C. Klaas, J. Hindhede, and J. T. Sørensen. 2006. The association between fetching for milking and dairy cows' behaviour at milking, and avoidance of human approach—An on-farm study in herds with automatic milking systems. *Livest. Sci.* 101:219–227. <https://doi.org/10.1016/j.livprodsci.2005.11.013>.
- Russell, A. L., L. V. Randall, J. Kaler, N. Eyre, and M. J. Green. 2023. Use of qualitative behavioural assessment to investigate affective states of housed dairy cows under different environmental conditions. *Front. Vet. Sci.* 10:1099170. <https://doi.org/10.3389/fvets.2023.1099170>.
- Santos, J. E. P., I. J. Lean, H. Golder, and E. Block. 2019. Meta-analysis of the effects of prepartum dietary cation-anion difference on performance and health of dairy cows. *J. Dairy Sci.* 102:2134–2154. <https://doi.org/10.3168/jds.2018-14628>.
- Schirmann, K., N. Chapinal, D. M. Weary, W. Heuwieser, and M. A. G. von Keyserlingk. 2012. Rumination and its relationship to feeding and lying behavior in Holstein dairy cows. *J. Dairy Sci.* 95:3212–3217. <https://doi.org/10.3168/jds.2011-4741>.
- Schwanke, A. J., K. M. Dancy, H. W. Neave, G. B. Penner, R. Bergeron, and T. J. DeVries. 2022. Effects of concentrate allowance and individual dairy cow personality traits on behavior and production of dairy cows milked in a free-traffic automated milking system. *J. Dairy Sci.* 105:6290–6306. <https://doi.org/10.3168/jds.2021-21657>.
- Schwanke, A. J., K. M. Dancy, H. W. Neave, G. B. Penner, R. Bergeron, and T. J. DeVries. 2024a. Effect of dairy cow personality traits and concentrate allowance on their response to training and adaptation to an automated milking system. *J. Dairy Sci.* 107:11446–11462. <https://doi.org/10.3168/jds.2024-25119>.
- Schwanke, A. J., H. W. Neave, G. B. Penner, R. Bergeron, and T. J. DeVries. 2024b. Flexible feeding: Dairy cow personality affects changes in feeding behavior and milk production under feed competition conditions. *J. Dairy Sci.* 107:2465–2482. <https://doi.org/10.3168/jds.2023-24063>.
- Vagnoni, D. B., and G. R. Oetzel. 1998. Effects of dietary cation-anion difference on the acid-base status of dry cows. *J. Dairy Sci.* 81:1643–1652. [https://doi.org/10.3168/jds.s0022-0302\(98\)75732-7](https://doi.org/10.3168/jds.s0022-0302(98)75732-7).
- Venjakob, P. L., L. Pieper, W. Heuwieser, and S. Borchardt. 2018. Association of postpartum hypocalcemia with early-lactation milk yield, reproductive performance, and culling in dairy cows. *J. Dairy Sci.* 101:9396–9405. <https://doi.org/10.3168/jds.2017-14202>.
- Vittorazzi, P. C. Jr., C. S. Takiya, A. T. Nunes, R. G. Chesini, M. Bugoni, G. G. Silva, T. B. P. Silva, M. S. S. Dias, N. T. S. Grigoletto, and F. P. Rennó. 2022. Feeding encapsulated pepper to dairy cows during the hot season improves performance without affecting core and skin temperature. *J. Dairy Sci.* 105:9542–9551. <https://doi.org/10.3168/jds.2022-22078>.
- Wilkens, M. R., C. D. Nelson, L. L. Hernandez, and J. A. A. McArt. 2020. Symposium review: Transition cow calcium homeostasis—Health effects of hypocalcemia and strategies for prevention. *J. Dairy Sci.* 103:2909–2927. <https://doi.org/10.3168/jds.2019-17268>.
- Zhang, J., L. Bu, Y. Liu, W. Huo, C. Xia, C. Pei, and Q. Liu. 2023. Dietary supplementation of sodium butyrate enhances lactation performance by promoting nutrient digestion and mammary gland development in dairy cows. *Anim. Nutr.* 15:137–148. <https://doi.org/10.1016/j.aninu.2023.08.008>.
- Zimpel, R., M. B. Poindexter, A. Vieira-Neto, E. Block, C. D. Nelson, C. R. Staples, W. W. Thatcher, and J. E. P. Santos. 2018. Effect of dietary cation-anion difference on acid-base status and dry matter intake in dry pregnant cows. *J. Dairy Sci.* 101:8461–8475. <https://doi.org/10.3168/jds.2018-14748>.

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